

WHARTON, TX, USA

WMO: 722351

Lat: 29.254N

Lon: 96.154W

Elev: 31

StdP: 100.96

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 12953

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.3	1.0	-7.8	2.0	7.5	-5.5	2.4	9.2	11.6	13.3	10.9	15.2	3.1	350	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.3	36.9	24.0	35.7	24.2	34.8	24.2	27.0	31.6	26.5	31.1	26.0	30.5	3.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.0	21.5	28.6	25.3	20.5	28.2	25.0	20.2	28.0	84.7	31.8	82.8	30.8	80.7	30.6	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.3	8.9	8.0	-3.4	39.0	2.0	1.4	-4.8	40.0	-6.0	40.9	-7.1	41.6	-8.5	42.7		
			-4.2	27.2	2.1	0.6	-5.8	27.7	-7.0	28.0	-8.2	28.4	-9.7	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	11.7	14.4	17.7	21.0	24.7	27.8	28.6	29.2	26.9	22.1	16.9	13.1	
	DBStd	7.19	5.08	5.42	4.54	3.59	2.77	1.59	1.50	1.73	2.23	4.10	5.18	5.27	
	HDD10.0	101	40	20	4	1	0	0	0	0	0	0	8	27	
	HDD18.3	703	212	130	67	16	1	0	0	0	0	16	86	176	
	CDD10.0	4198	93	143	244	331	457	534	575	594	508	377	217	125	
	CDD18.3	1758	6	20	48	97	200	284	317	336	258	133	43	15	
	CDH23.3	17191	22	71	234	692	1791	3037	3487	3822	2449	1235	296	57	
	CDH26.7	7679	0	3	23	150	656	1469	1731	2013	1123	461	48	2	
Wind	WSAvg	3.1	3.6	3.9	3.9	4.0	3.5	2.7	2.3	2.2	2.2	2.6	3.0	3.4	
Precipitation	PrecAvg	1109	81	66	51	62	112	125	84	85	153	119	76	75	
	PrecMax	2007	215	183	145	177	350	408	224	185	358	422	211	184	
	PrecMin	628	5	7	6	0	11	8	8	6	23	4	3	16	
	PrecStd	300	52	41	42	47	79	105	60	52	95	104	52	45	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.5	27.1	28.7	31.8	34.0	37.3	37.2	38.7	37.0	33.3	29.9	26.7	
		MCWB	18.2	18.8	19.1	21.0	22.6	22.9	23.9	25.3	22.4	22.4	21.4	20.6	
	2%	DB	23.7	25.4	27.3	30.0	32.6	35.2	36.0	37.4	35.0	32.2	27.9	25.0	
		MCWB	17.9	19.3	19.8	20.6	22.7	24.0	24.2	24.5	23.7	22.9	21.2	20.4	
	5%	DB	22.2	23.9	26.0	28.1	31.6	34.0	35.0	36.1	33.6	31.0	26.8	23.3	
		MCWB	18.1	19.0	19.6	20.6	22.7	24.2	24.4	24.5	23.6	22.6	20.5	19.7	
	10%	DB	20.6	22.4	24.7	27.2	30.4	32.8	33.7	34.9	32.5	29.7	25.0	22.1	
		MCWB	17.2	18.8	19.0	20.4	22.7	24.2	24.5	24.6	23.8	21.8	20.0	19.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.3	21.7	22.4	24.3	25.5	27.0	27.4	27.8	27.3	26.3	24.0	22.9
			MCDB	22.2	24.0	25.9	27.8	29.6	31.2	32.0	32.8	31.4	29.8	27.1	24.5
		2%	WB	20.3	21.1	21.7	23.5	24.9	26.4	26.8	27.1	26.5	25.4	23.0	21.9
			MCDB	22.0	23.1	24.7	27.0	29.2	31.0	31.0	32.1	30.8	28.7	25.5	23.6
5%		WB	19.2	20.6	21.1	22.8	24.4	25.8	26.2	26.6	25.8	24.4	22.2	20.9	
		MCDB	21.6	22.6	23.6	26.2	28.7	30.3	30.3	31.6	29.9	28.1	25.0	22.7	
10%		WB	17.8	19.9	20.5	22.1	24.0	25.4	25.7	26.0	25.2	23.5	21.2	19.8	
		MCDB	19.5	22.1	23.0	25.3	28.1	29.8	30.1	30.9	29.3	27.3	24.0	21.7	
Mean Daily Temperature Range	5% DB	MDBR	11.3	10.8	10.8	11.0	10.2	10.3	10.5	11.3	10.7	12.5	11.7	10.8	
		MCDBR	12.3	10.9	10.6	11.5	11.2	12.0	12.3	12.9	12.3	12.4	11.8	11.4	
		MCWBR	7.9	6.4	5.3	5.2	3.7	3.3	3.4	3.3	3.9	4.9	6.2	7.7	
	5% WB	MCDBR	10.4	9.0	8.6	8.6	8.9	9.7	10.0	11.1	10.1	10.0	10.0	10.1	
		MCWBR	8.2	6.6	5.2	4.7	3.4	3.3	3.4	3.5	3.9	5.0	6.3	8.1	
	Clear-Sky Solar Irradiance	taub	0.344	0.356	0.377	0.431	0.483	0.457	0.485	0.472	0.447	0.383	0.375	0.346	
taud		2.482	2.457	2.389	2.244	2.134	2.267	2.204	2.242	2.277	2.444	2.411	2.485		
Ebn at Noon		893	911	910	867	820	837	812	821	830	870	847	871		
Edh at Noon		93	103	117	140	157	137	145	139	130	103	99	89		
All-Sky Solar Radiation	RadAvg	2.89	3.30	4.45	5.31	5.94	6.34	6.25	5.96	5.06	4.38	3.29	2.59		
	RadStd	0.30	0.57	0.39	0.44	0.41	0.45	0.57	0.39	0.50	0.49	0.36	0.32		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page